

MAT1011 Homework 1

Instructions. Show the essential reasoning in every problem. Unless an ε - δ proof is explicitly requested, you may use the standard limit laws and the squeeze theorem. You may use $|\sin t| \leq |t|$ and the density of rational and irrational numbers without proof. Do not use derivatives, L'Hopital's rule, or the sequential criterion for limits. Functions are real-valued and are understood to be defined on an open interval about the limit point, except possibly at that point.

1. COMPULSORY EXERCISES

Question 1. Find the following limits. Simplify the expressions algebraically before applying limit laws.

(a) $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 4}.$

(b) $\lim_{x \rightarrow 0} \frac{\sqrt{1+2x} - \sqrt{1-x}}{x}.$

(c) $\lim_{x \rightarrow 0} \frac{x^2 \sin(1/x)}{\sqrt{1+x} - 1}.$

(d) Let $m, n \geq 1$ be fixed integers. Find $\lim_{x \rightarrow 1} \frac{x^n - 1}{x^m - 1}.$

Question 2. Suppose

$$\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = 1.$$

Work directly from the ε - δ definition of the given limit.

(a) Prove that

$$\lim_{x \rightarrow 0} \frac{f(x)}{x} = 0 \quad \text{and} \quad \lim_{x \rightarrow 0} f(x) = 0.$$

(b) Show first that $f(x) > 0$ for all sufficiently small $x \neq 0$. Then prove that

$$\lim_{x \rightarrow 0} \frac{\sqrt{f(x)}}{|x|} = 1.$$

Question 3. This problem concerns the logical form of the definition of a limit.

(a) Show by example that the following statement is false:

“The number L is the limit of $f(x)$ as $x \rightarrow c$ if, for every $\varepsilon > 0$ and every $\delta > 0$, there exists an x in the domain of f such that $0 < |x - c| < \delta$ and $|f(x) - L| < \varepsilon$.”

Compare its quantifier pattern with the quantifier pattern in the correct definition of a limit.

(b) Write the exact negation of $\lim_{x \rightarrow c} f(x) = L$. Then use that negation to prove that $\lim_{x \rightarrow 0} \sin(1/x)$ does not exist as a finite real number.

Question 4. Find each limit and then prove your answer using the ε - δ definition. In part (c), you may use the angle-sum identities and $|\sin t| \leq |t|$.

(a) $\lim_{x \rightarrow 1} \frac{x^2 - 1}{2x^2 - x - 1}.$

(b) Let $x_0 \neq 0$. Find $\lim_{x \rightarrow x_0} \frac{1}{x}.$

(c) Find

$$\lim_{x \rightarrow x_0} \sin x \quad \text{and} \quad \lim_{x \rightarrow x_0} \cos x.$$

(d) Let $|x_0| < 1$. Find $\lim_{x \rightarrow x_0} \sqrt{1 - x^2}.$

(e) Here $[y]$ denotes the greatest integer not exceeding y . Find $\lim_{x \rightarrow 0} x \left[\frac{1}{x} \right].$

Question 5. Prove the following basic properties of limits directly from the ε - δ definition.

(a) A finite limit, if it exists, is unique.

(b) Suppose that $f(x) = g(x)$ whenever $0 < |x - c| < r$ for some $r > 0$. Prove that $\lim_{x \rightarrow c} f(x) = L$ if and only if $\lim_{x \rightarrow c} g(x) = L$. Deduce that changing the value of a function at $x = c$ does not change its limit as $x \rightarrow c$.

(c) If $\lim_{x \rightarrow c} f(x) = A$, prove that $\lim_{x \rightarrow c} |f(x)| = |A|$. For which values of A is the converse valid for all such functions f ? Justify your answer.

Question 6. Recall the Dirichlet function

$$D(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}, \end{cases} \quad F(x) = xD(x).$$

(a) Prove that $\lim_{x \rightarrow 0} F(x) = 0$.

(b) Prove that $\lim_{x \rightarrow x_0} F(x)$ does not exist at any $x_0 \neq 0$.

Question 7. Suppose that

$$\lim_{x \rightarrow c} f(x) = A, \quad \lim_{x \rightarrow c} g(x) = B,$$

and that $f(x) < g(x)$ whenever $0 < |x - c| < r$ for some $r > 0$. Prove directly from the ε - δ definition that $A \leq B$. Must $A < B$ hold? Justify your answer with an example.

2. EXTRA EXERCISES

Question 8. Suppose that

$$\lim_{x \rightarrow c} (f(x) + g(x)) = 0, \quad \lim_{x \rightarrow c} f(x)g(x) = 0.$$

Prove that $\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} g(x) = 0$. Can either assumption be omitted? Give a counterexample for each omission.

Question 9.

(a) Prove that

$$\lim_{x \rightarrow x_0} f(x) = A \iff \lim_{h \rightarrow 0} f(x_0 + h) = A.$$

(b) Let f be defined for $0 < |x| < r$, where $r > 0$. Prove directly from the ε - δ definition that

$$\lim_{x \rightarrow 0} f(x) = A \implies \lim_{t \rightarrow 0} f(t^2) = A.$$

Is the converse true? Give a counterexample and justify it.

3. THOMAS TEXTBOOK EXERCISES

Use the same edition of Thomas' textbook adopted for this course.

Chapter 1. Limits and Differentiation

Section 1.1. Limits and Continuity

§1.1.1. Intuitive Introduction of Limit

Exercises 2.2: 22, 34, 42.

§1.1.2. The Limit Laws and §1.1.3. Mathematical Definition of Limit

Exercises 2.3: 44, 50, 54.